

ADDITIONAL PROBLEM SHEET 3

Topics covered: Chapters 6–7.

1. Travelling waves (Exam section A type)

Consider the viscous Berger equation for a scalar density u :

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} - \nu \frac{\partial^2 u}{\partial x^2} = 0, \quad (1)$$

where ν is a positive constant.

- (a) Assume a travelling wave solution $u(x, t) = v(z)$, where $z = x - ct$ and c is a constant wavespeed. Show that eq. (1) reduces to

$$-cv' + vv' - \nu v'' = 0, \quad (2)$$

where $v' = dv/dz$.

- (b) Show that this implies

$$v' = \frac{1}{2\nu} (v^2 - 2cv - 2B), \quad (3)$$

where B is a constant of integration.

- (c) The right-hand side of eq. (3) can be written as

$$\frac{1}{2\nu} (v - v_1)(v - v_2), \quad \text{where } v_1 = c - \sqrt{c^2 + 2B}, \quad v_2 = c + \sqrt{c^2 + 2B} \quad (4)$$

(assuming $c^2 > 2B$). Integrate eq. (3) to show

$$v(x) = \frac{v_2 - v_1 e^{K(z)}}{1 - e^{K(z)}}, \quad K(z) = (v_2 - v_1) \left(\frac{z}{2\nu} + C \right). \quad (5)$$

with C a constant of integration.

2. Travelling waves (Exam section A, 2017)

The Korteweg–de Vries (KdV) equation,

$$\frac{\partial u}{\partial t} + 6u \frac{\partial u}{\partial x} + \frac{\partial^3 u}{\partial x^3} = 0, \quad (6)$$

has, in recent years, been used as a model for the vibration of biopolymers.

- (a) If we seek a travelling wave solution $u = v(z)$, $z = x - ct$, show that the KdV equation reduces to

$$-c \frac{dv}{dz} + 6v \frac{dv}{dz} + \frac{d^3 v}{dz^3} = 0. \quad (7)$$

(b) Writing $dv/dz = v'$, deduce that

$$\frac{(v')^2}{2} = -v^3 + \frac{c}{2}v^2 + Av + B, \quad (8)$$

where A and B are constants of integration.

(c) Assume that $v, v', v'' \rightarrow 0$ at the same z value. Use this assumption to show that

$$u(x, t) = \frac{c}{2} \operatorname{sech}^2 \left[\pm \frac{\sqrt{c}}{2} (x - ct + D) \right], \quad (9)$$

where D is a constant of integration. You are advised to use the substitutions $v = \frac{1}{2}c \operatorname{sech}^2(\theta)$ and $v\sqrt{c-2v} = \frac{1}{2}c^{3/2} \operatorname{sech}^2(\theta) \tanh(\theta)$ to integrate.

3. Chemotaxis-based problem, feat. travelling waves (Exam section B type)

Consider the following system of diffusive equations based on a model for bacterial chemotaxis:

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2} - \frac{\partial}{\partial x} \left(u \frac{\partial v}{\partial x} \right), \quad (10)$$

$$\frac{\partial v}{\partial t} = -ku. \quad (11)$$

The bacteria $u(x, t)$ increase their population due to the presence of a (chemical) food source $v(x, t)$. The bacteria move via diffusion, the first term of the first equation, and chemotaxis, represented by the second term. The parameters D and k are assumed constant.

We seek travelling wave solutions with the variables x, t related via $z = x - ct$, depending on the spatial gradient of v .

(a) If we assume that when $u \rightarrow 0$, $du/dz \rightarrow 0$ and $u(0) = 1$, show that $u(x, t)$ takes the form

$$u(x, t) = \frac{c^2}{k - (k - c^2)e^{c(x-ct)/D}}. \quad (12)$$

(b) Assuming both k and c are positive constants, under what circumstances is this model physically valid? Describe the shape of the solution for fixed t as function of x .

(c) What happens to this distribution as a function of time and what does this suggest about the eventual behaviour of the function $v(x, t)$ as $t \rightarrow \infty$?

(d) What happens to the distribution, again at fixed t , in the limit $D \rightarrow 0$? What is the physical significance of this limit?

4. PDE stability (Exam section A type)

PDE stability will return in Epiphany term, so if this seems alien to you before you see it again, feel free to hold off.

Consider the following partial differential equation:

$$\frac{\partial^2 y}{\partial x^2} + y^2 \frac{\partial y}{\partial x} + C(1 - y) = \frac{\partial y}{\partial t}. \quad (13)$$

We assume $x \in [0, L]$ and that y satisfies $y(0, t) = y(L, t) = \alpha$, for some constant α .

- (a) Find the spatially homogeneous equilibrium of eq. (13).
- (b) Hence explain why the constant α must equal 1.
- (c) Find the domain of C for which this equilibrium is stable and state the form of the linearised solutions.
- (d) If we had instead chosen the boundary conditions

$$\frac{\partial y}{\partial x}(0, t) = \frac{\partial y}{\partial x}(L, t) = 0, \quad (14)$$

what would be the requirement on C for a stable spatially homogeneous equilibrium?

5. PDE stability (Exam section A type)

Consider the following diffusion-type model for a scalar density u :

$$\frac{\partial u}{\partial t} = D_1 \frac{\partial^2 u}{\partial x^2} + D_2 \frac{\partial^4 u}{\partial x^4}, \quad u(0) = 0, \quad u(L) = 0, \quad (15)$$

where $D_1, D_2 \geq 0$ are constants and $x \in [0, L]$.

- (a) Find the homogeneous equilibrium of this system.
- (b) Consider the case $D_1 > 0$ and $D_2 = 0$. Show that this homogeneous equilibrium is asymptotically stable, whatever the value of the domain length, L .
- (c) Consider the case $D_1, D_2 > 0$. Find an expression for the minimum integer wavemode n which is unstable for a given length, L .
- (d) *[A part A question would have stopped there, but here's one more question for the sake of revision]*

Consider a similar model,

$$\frac{\partial u}{\partial t} = D_1 \frac{\partial^2 u}{\partial x^2} + D_2 \frac{\partial u}{\partial x}, \quad u(0) = 0, \quad u(L) = 0. \quad (16)$$

Assess the stability of the homogeneous equilibrium $u = 0$.

6. Supported Euler buckling (Exam section A, 2017)

Consider an Euler beam of length $L = 1$ under a load N . The beam is placed in a spongy medium which will resist its bending. Modelling the medium as having a resistive spring constant μ , the modified beam equation for the deflection w , as a function of arclength s and time t , takes the form

$$\frac{\partial^4 w}{\partial s^4} + N \frac{\partial^2 w}{\partial s^2} + \mu w + \rho \frac{\partial^2 w}{\partial t^2} = 0, \quad (17)$$

with ρ , N and μ positive constants. This has been used as a model for the microtubule structures which embed in cell cytoplasm to provide rigidity to the cell structure. There is an equilibrium $w(s) = 0, \forall s \in [0, 1]$.

- (a) Show using a linear stability analysis that the condition required for stability of this equilibrium is

$$k^4 - Nk^2 + \mu > 0, \tag{18}$$

where k is the wavenumber of the perturbation.

- (b) Calculate the critical load N at which the beam becomes unstable and gives way, on the assumption that $N^2 \geq 4\mu$ and $\mu = 81\pi^4$. You are advised that this does *not* occur for the $n = 1$ mode.